

# 8.4.5

EE25BTECH11002 - Achat Parth Kalpesh

## Question:

An ellipse is drawn by taking a diameter of the circle  $(x - 1)^2 + y^2 = 1$  as its semi minor axis and a diameter of the circle  $x^2 + (y - 2)^2 = 4$  as semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is

$$1) 4x^2 + y^2 = 4$$

$$2) x^2 + 4y^2 = 8$$

$$3) 4x^2 + y^2 = 8$$

$$4) x^2 + 4y^2 = 1$$

## Solution:

The standard equation of a circle is given as

$$(\mathbf{x} - \mathbf{c})^T (\mathbf{x} - \mathbf{c}) = r^2 \quad (4.1)$$

Given two circles are

$$(\mathbf{x} - \mathbf{c}_1)^T (\mathbf{x} - \mathbf{c}_1) = 1 \quad (4.2)$$

$$(\mathbf{x} - \mathbf{c}_2)^T (\mathbf{x} - \mathbf{c}_2) = 4 \quad (4.3)$$

The centers and radii are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad r_1 = 1 \quad (4.4)$$

$$\mathbf{c}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \quad r_2 = 2 \quad (4.5)$$

Verifying that the origin lies on both circles:

$$(\mathbf{0} - \mathbf{c}_1)^T (\mathbf{0} - \mathbf{c}_1) = 1 = r_1^2 \quad (4.6)$$

$$(\mathbf{0} - \mathbf{c}_2)^T (\mathbf{0} - \mathbf{c}_2) = 4 = r_2^2 \quad (4.7)$$

Thus, the diameters of both circles passing through the origin are along the directions

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ (along X-axis)} \quad (4.8)$$

$$\mathbf{c}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \text{ (along Y-axis)} \quad (4.9)$$

Each circle's diameter length is  $2r$ . Therefore, the ellipse's semi-axes are equal to the respective radii:

$$b = r_1 = 1 \quad \text{(semi-minor axis)} \quad (4.10)$$

$$a = r_2 = 2 \quad \text{(semi-major axis)} \quad (4.11)$$

The standard equation of an ellipse centered at the origin with coordinate axes as its axes is

$$\mathbf{x}^T A \mathbf{x} = 1 \quad (4.12)$$

where

$$A = \begin{pmatrix} \frac{1}{b^2} & 0 \\ 0 & \frac{1}{a^2} \end{pmatrix} \quad (4.13)$$

Substituting  $a = 2$ ,  $b = 1$ ,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \quad (4.14)$$

Hence,

$$\mathbf{x}^T \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \mathbf{x} = 1 \quad (4.15)$$

Multiplying throughout by 4 gives

$$\mathbf{x}^T \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} = 4 \quad (4.16)$$

or equivalently,

$$4x^2 + y^2 = 4 \quad (4.17)$$

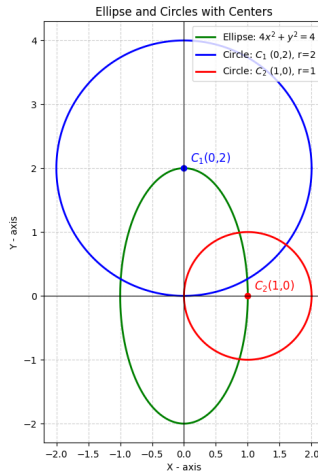


Fig. 4.1: Ellipse and Circles